

Technical University of Munich, Munich, Germany

August 1, 2026

Dear PhD Committee,

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As a passionate AI Data Science Engineer with a strong background in machine learning and programming, I am excited to apply for the Research Associate Position in Contrastive Learning and GeoAI at the Technical University of Munich (TUM). With my Master's degree in related fields and proficiency in Python, PyTorch, and TensorFlow, I believe I have the ideal skillset to contribute to this innovative research project.

Throughout my academic journey, I have been fascinated by the potential of machine learning to transform geospatial data analysis. My experience with developing medical digital twins has given me a unique perspective on the importance of sampling strategies for geospatial datasets. Specifically, my work on *Medical_MambaTS, a linear-time state space model for sub-millisecond bedside inference, demonstrates my scale georeferenced data.*

I am particularly drawn to this PhD position because it allows me to leverage my knowledge of contrastive learning and apply it to the field of GeoAI. I am impressed by the Professorship Big Geospatial Data Management's emphasis on methods of distributed computing, machine learning, and image analysis, which aligns with my research interests.

In conclusion, I believe that my skills and experience make me an excellent candidate for this Research Associate Position. I look forward to contributing to the development of innovative sampling strategies for geospatial datasets and exploring the applications of contrastive learning in GeoAI.

Sincerely,

Ismail Aissaoui

Ismail Aissaoui